A corrected pole picture for a shortcut into an absorbing site on a one-dimensional ring

Abstract

This note records the corrected stochastic interpretation of a shortcut from a ring site into an absorbing target. The key point is that, after the absorbing site has been removed from the transient state space, the shortcut is a killing channel rather than a probability-conserving transfer between transient sites. This changes the sign in the rank-one resolvent denominator, invalidates the old pole picture, and separates the asymptotic tail question from finite-time visual bimodality.

1 Role of this note

This document has a different job from the in-place redline audit of Luca's original manuscript. The redline audit is author-facing: it preserves the original text and marks the precise wrong steps. This note is the cleaned research-facing version. It can sit inside the Valley-K ring material as the corrected reference for the absorbing shortcut calculation.

For the broader one-dimensional ring programme, the calculation is useful in three ways. First, it gives a small exactly-checkable model in which shortcut strength changes the first-passage tail. Second, it provides a warning about rank-one updates: a shortcut into an absorbing target is not algebraically the same object as a shortcut into another live state. Third, it gives a clean test case for separating asymptotic tail decay from finite-time shape effects such as apparent bimodality.

2 Transient-space model

Let the ring have N sites and let v be absorbing. The transient state space is the set of all sites except v. Let P_0 be the transient transition matrix for the lazy nearest-neighbour walk with absorbing target already removed. Suppose a shortcut starts at u and lands at v. If β is the shortcut strength and q is the move probability, define

$$\lambda = \beta(1-q).$$

Because the endpoint v is absorbing, the shortcut removes probability from the transient state space. Therefore the correct transient matrix is

$$P_\beta = P_0 - \lambda|u\rangle\langle u|. \tag{1}$$

This is the sign that the original pole diagram missed.

3 Corrected rank-one resolvent

Let

$$W(z) = (I - zP_0)^{-1}$$

be the baseline transient resolvent. The corrected shortcut resolvent is

$$(I - zP_\beta)^{-1} = W(z) - \frac{z\lambda\, W(z)|u\rangle\langle u|W(z)}{1 + z\lambda W_{u,u}(z)}. \tag{2}$$

The plus sign in the denominator is forced by $P_\beta = P_0 - \lambda|u\rangle\langle u|$. Any pole equation derived from

$$1 - z\lambda W_{u,u}(z)$$

belongs to the wrong probability-conserving shortcut model. It should not be used for the shortcut-to-absorbing-target calculation.

4 Symmetric ring and pole geometry

In the symmetric case let $N = 2L$, $v = 0$, and $u = L$. Write

$$s = \frac{1}{z}, \qquad y = \frac{s - 1 + q}{q}, \qquad a = \frac{q}{\beta(1-q)}.$$

The Chebyshev form of the corrected pole denominator can be written as

$$D_+(s) = a + \Phi_L(s), \qquad \Phi_L(s) = T_L(y)U_{L-1}(y), \tag{3}$$

where T_L and U_{L-1} are Chebyshev polynomials of the first and second kind. Thus the corrected zero condition is

$$\Phi_L(s) = -a. \tag{4}$$

The old sign instead asks for $\Phi_L(s) = +a$. That is the wrong pole geometry for the stochastic shortcut-to-absorbing-target interpretation.

5 Tail asymptotics and finite-time shape

The first-passage tail is controlled by the spectral radius of the killed transient matrix P_β, or equivalently by the corrected pole of the generating function with largest modulus in z. In $s = 1/z$ variables this dominant value is the largest relevant decay factor below one. The tail has the form

$$\Pr(T = t) \sim C\,\rho(P_\beta)^{t-1}$$

for the appropriate residue C, provided the dominant pole is simple.

This is not the same question as finite-time visual bimodality. A distribution can have two visible finite-time scales while still having a single asymptotic tail. Conversely, a change in the corrected tail gap does not by itself prove a visual bimodality threshold. Finite-time bimodality should be checked from the full coefficient sequence or by direct simulation of the killed chain.

6 Implication for the Valley-K ring work

This corrected note should be treated as a local analytic guardrail for the one-dimensional ring part of the Valley-K project. It says exactly which shortcut perturbation is being studied, supplies the corrected pole equation, and gives a reproducible figure based on the same killed-chain model. The note is most useful as a technical appendix or companion to numerical experiments: it can calibrate simulations, check finite-time first-passage plots, and prevent the invalid threshold from being used as an explanation of ring bimodality.

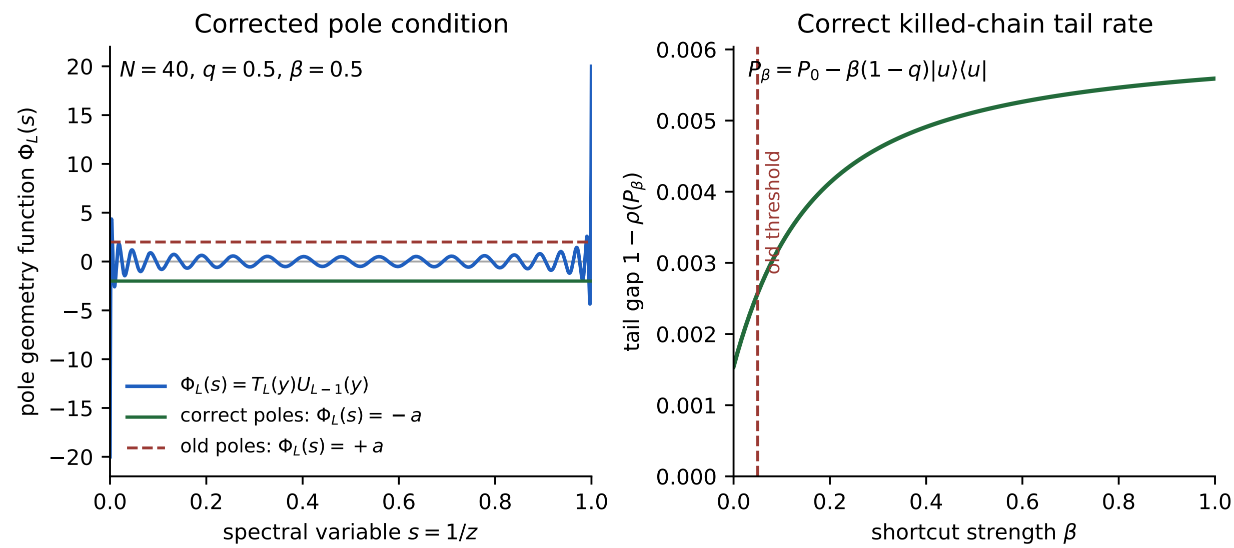

Figure 1: Corrected pole geometry and tail-rate diagnostic for the symmetric absorbing shortcut on a ring. Left: the corrected pole equation uses $\Phi_L(s) = -a$, whereas the old manuscript used the opposite sign. Right: the tail gap $1 - \rho(P_\beta)$, computed directly from the killed transient matrix in (1), varies smoothly with shortcut strength; the old threshold is shown only as a marker for the invalid sign-based criterion.